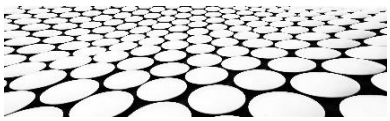
CSI4142 Fundamentals of Data Science



ASSIGNMENT 4

Unsupervised Learning
Clustering and Recommendations

GOALS

The overall goal of this assignment is to become familiar with both clustering methods and recommendation systems.

At the end of this assignment, you will have:

- Reviewed your Python skills, as the project MUST be done in Python
- Explored Kaggle, an amazing resource of challenges and datasets
- Experimented with different similarity measures
- Experimented with 2 methods of clustering: KMeans and DBSCAN
- Experimented with content-based recommendation
- Experimented with collaborative filtering recommendation (model-based)
- Documented everything, in Jupyter Notebooks, in a way to make your analysis understandable and reproducible.

SUBMISSION INFORMATION

- Deadline:
  - Submission of your notebook: **Tuesday, April 7th, midnight**
  - Grace period (no penalty) is until 12:30… after that it's 20% per day off.
- Groups:
  - **You are expected to form groups of 2 and do a single submission per group**.
  - You need to register your group in Brightspace. As we are allowing groups to change from one assignment to another, <u>you have to register again your group.</u> This time as a A4 group.
  - It is not possible to work alone, and to enforce such policy, *a penalty of 20% will be imposed on solo groups.*
- Where to submit:
  - Your submission must be done in Brightspace in Assignment section (Assignment 4)
- Submission format:
  - <u>Only Jupyter Notebook files will be accepted to show your programs</u>
  - For the dataset used in your analysis, your code MUST contain a link to the dataset so the TAS do NOT need to download any data.
- Demo/Presentation (**mandatory**):
  - You must select a time (a set of timeslots will be provided) to meet with one of the TAs to present your work.
  - Both students must be present during the presentation. If one student is missing, that student will have a 20% penalty on the entire assignment. The student who missed will not be allowed to reschedule as the demo is already done.
  - A group that has a valid reason can contact the TA to reschedule without penalty, but it must be done at least 24 hours before your time slot, and there must still be available timeslots. A group cannot reschedule twice.
  - If you miss your time slot, it is 20% penalty on the entire assignment. To not have zero, you should reschedule within 24 hours of your missed slot, and while there are still available timeslots.
  - Any no-show will be given zero on the assignment. **Even if the demo part counts for 12 marks out of 60, not doing the demo results in zero for the entire assignment.**

PLEASE NOTE:

If the corrector cannot run your code, the mark will be zero for the entire assignment. It is your responsibility to test that the code cells in your Notebook are executable. The <u>most frequent problem</u> is that the dataset is not readable (file not found). Please make sure the dataset is either read directly from a public place (a dataset repository) or read from one of your shared repository (accessible to anyone) so the TA can run your notebook without any data download.

PROGRAMMING REQUIREMENTS

1. **Choose a dataset**

As you know, the Kaggle site is a very good site to explore as it contains datasets for many tasks in data science and AI. There are also other sites for datasets if you wish to further explore. ***You must choose one dataset that will allow you to realize the diverse studies described in Section 3.*** Your chosen dataset cannot be a dataset previously used in another assignment.

Dataset Suggestions

1. The Movies Dataset
   - Link : The Movies Dataset
2. Amazon apparels data review
   - Link : Apparel reviews

In Movies dataset, there are two important files to focus on:

- *movies_metadata.csv* contains metadata about the movies. Some of the values are in JSON format, but others (e.g. budget, release_date) are not. You would use this file for the studies 1 to 3 (similarity, clustering, content-based recommendation)
- *ratings.csv* or *ratings_small.csv*. You can work with *ratings_small.csv* if you find its size more manageable. You will use this file for study 4 on collaborative filtering. It contains individual ratings to build the utility matrix. The movieId in that file can be used to retrieve movie metadata in the movies_metadata file.

In Amazon apparels dataset, there are also two files:

- *Amazon-clothing-info.csv* containing metadata about clothing. There are many missing values, but still the dataset contains some features (e.g color, price) on which it would be possible to do similarity, clustering and content-based recommendations. It's a messy dataset, but perhaps more representative of "real data".
- *Clothing-Reviews.csv* gives a review_score and a review_userId which you need for the collaborative filtering. The feature asin will allow you to connect the review to its metadata.

The datasets above are only suggestions that would allow you to realize the four required studies. If you prefer, you can pick a different dataset, as long as it allows you to perform the different studies.

If you select a different dataset, it must contain metadata and individual ratings. These will usually be provided in 2 files. For studies 1 to 3, you need the metadata, and for study 4, you absolutely need individual user ratings. The 2 datasets suggested contain some metadata to describe the items (movies, apparels) as well as *individual* ratings of the items by users. By *individual ratings*, it means that the dataset includes ratings from different users which is different from an attribute on each item showing an agglomerated score or popularity.

2. Prepare the data

With the experience that you now have from the first 3 assignments, you can clean the data and prepare it for the studies. You can do EDA, data cleaning (validation, outlier detection, missing value imputation) and feature engineering (to create other features that could be useful).

The requirements for this section are very open and let you judge/decide what is required. Just show a few tests (at least 3) that you did or graphs that you generated in this exploration.

3. Studies

You are asked to program different studies using the same recommendation dataset***. I will use the Movie dataset as an example when describing the studies.***

I call them simply "studies" instead of "empirical studies" because we are not following a systematic empirical approach with proper evaluation (except for Study 4). We are rather doing simulations to explore different ideas.

The purpose of each study is to make you better understand some concepts seen in class relating to similarity measures, clustering and recommendation systems.

As this is the last assignment, the descriptions and requirements for each study are a bit more open. I encourage you to try different things and report on your findings. You are required to show the studies in order (Study 1 to 4) in your notebook. That will make it easier for the demonstration and the marking. But, the exact content of each cell is not specified, so you can present the results in a way you think will best emphasize your effort and results.

Study 1 – Similarity measures

For this study, you will program similarity measures on different attributes independently.

Pick a subset of 5 attributes (e.g. Genre, Revenue, Runtime, Title, Budget) in your dataset. The subset of attributes should allow you to explore at least 3 different similarity measures (e.g. Jaccard, Hamming, Cosine, Euclidian, Manhattan). You should also explore an additional similarity measure on textual data. For example, my choice of Title as attribute will require a similarity measure that could compare "Toy Story" with "Sad Story". *In class, I mentioned that you would have to explore these string comparison measures and hinted at Edit Distance as a possibility.*

Please note: If the dataset that you chose does not have 5 different metadata attributes to test, then try different similarity measures on the same ones. Please explain this in your report.

Simulate 5 user requests to show the results of your 5 similarity measures. For each request, show the Top 10 results (ranked on the criteria that you choose). For the movies, I could decide to rank on popularity or recency.

For example, using my similarity measures on the movie dataset, I could answer the following requests and show the Top 10 results.

- Show me movies of the same genre as "Toy Story" *(similarity on genre - categorical)*
- Show me movies with similar revenue to "Titanic" *(similarity on revenue - numerical)*
- Show me movies with similar length as "Apollo 13" *(similarity on duration - numerical)*
- Show me movies with similar title to "Fight Club" *(similarity on title – text based)*
- Show me movies with similar budget to "Matrix" *(similarity on budget – numerical)*

When showing the results, you can decide what metadata you think is relevant to show. For movies, perhaps my 10 results will include title, genre, runtime and popularity. You can decide to output different information.

Study 2 – Clustering algorithms

The purpose of this study is to test 2 different clustering algorithms. We saw KMeans and DBSCAN in class. You are required to test those two algorithms.

For each clustering algorithm, try at least a few variations on the hyper-parameters. For KMeans, test at least 2 values for k(1), and for DBSCAN, test at least 2 values for eps and min_samples.

Clustering is based on similarity. You are asked to combine at 2 attributes (from Study 1) for the similarities you use for clustering. Try at least 2 different combinations of 2 attributes. Limiting the similarity to be on 2 attributes will make it easier to do visual inspection on the resulting clusters.

Show your visual inspection results(2) for your combinations of attributes. For example, for the movie dataset, I can cluster and visualize results on (Revenue versus Budget) and (Runtime versus Budget), and that for each of the clustering algorithm.

Discuss which clustering algorithm (KMeans versus DBSCAN) seems more appropriate based on your visual inspection. Beyond visual inspection, you can also explore the cluster metrics we discussed in class (Cluster magnitude and cardinality), but that is optional.

Notes:

(1) You can also try to find the best value for K using the method shown during the lecture (trying k from small to large and measuring the cumulated magnitude over all clusters). That is optional.

(2) There is a lot of code online to perform cluster visualization in Python, such as shown in the Demo of DBSCAN clustering algorithm (link) in sklearn.

Study 3 – Content-Based Recommendation System

For this study, develop 2 similarity heuristics each taking a subset of features. ***A heuristic is a domain informed strategy.*** For example, I could program one similarity heuristic that uses Jaccard similarity on movie genre combined with an Euclidian distance on runtime and popularity. This heuristic would simulate the fact that probably people would want to watch new movies if they are of a similar genre, duration and popularity.

These heuristics can be any combination of the similarity measures developed in Study 1.

Your ***content-based recommendation system will make use of these heuristics to provide movie recommendations***. We did talk in class how content-based recommendation is subjective, so that subjectivity is reflected in your heuristics.

Simulate a few requests (at least 3) to show the retrieved recommendations based on your 2 different heuristics. For each request and each heuristic, show the Top 10 results (ranked on the criteria that you choose and displaying the information that you want).

For example, on the movie dataset, I could answer the following requests.

- Show me movies similar to "Toy Story"
- Show me movies similar to "Apollo 13"
- Show me movies similar to "Fight Club"

And for each request, I can decide to rank the top 10 based on popularity or recency and for each movie, I will show title, genre and popularity. You decide how you want to do this part.

Since you have 2 heuristics and 3 requests, you can compare results. Did your 2 heuristics provide similar top 10 results or not? Does it seem (qualitatively) that one heuristic suggests more relevant movies than the other?

You are not asked to program any metrics (e.g. Precision@k or MRR which will be done in Study 4) to perform a quantitative evaluation. Rather, you are asked to do a qualitative evaluation, meaning report your insights based on result inspection.

Study 4 – Collaborative Filtering Recommendation System

For this study, you will program a Recommendation System using model-based collaborative filtering.

The core of this approach is to first build the R_{ui} matrix (Utility matrix). The movie dataset contains a file with the movies metadata and another file with user ratings for various movies. The user ratings are used for building the Utility matrix. The collaborative filtering approach does not rely on metadata.

You will perform Matrix Factorization to decompose the R_{ui} matrix and approximate it with matrices P and Q (as seen in class). You can find code online to do so (e.g. Medium article "Recommender System – Matrix Factorization"). You can test different values for the latent dimensions.

Once you have matrices P and Q, you can predict any movie rating for any user. Those predictions can be used to establish the top rated movies that a user hasn't seen yet and would be likely to enjoy.

Simulate 3 requests of unseen movies, and for each user, find their top X rated movies and show the results.

As ***qualitative evaluation*** *(optional)*, you can let your recommendation system make recommendations and determine if they seem to make sense (e.g. seem likely based on movies they have rated highly).

For a ***quantitative evaluation***, you will simulate unrated data with rated one. To be able to evaluate your approach, remove 10% of the data, meaning 10% of the user ratings of various items. *Make sure that the 10% of ratings removed are sampled over all users.* This 10% of data becomes your Gold Standard (GS).

You will be able to compare the prediction scores (obtained by multiplying P and Q) with the gold standard ratings. You are asked to code different evaluation scores applied to the 10% saved data.

- Evaluation 1: Use a measure (e.g. MSE) to compare predicted scores against gold standard scores.
- Evaluation 2: In GS, transform the scores so that they become binary (relevant/non-relevant). For example, movies with scores 4 and 5 would be considered relevant and movies with scores 0 to 3 would be irrelevant. Now that all movies in GS are tagged as relevant/non-relevant, you can calculate P@5 and P@10, as well as the MRR.

Present and discuss the results.

☆☆☆ EVALUATION (60 points)

Jupyter Notebook (48 points)

- Overall effort in the report (5 points)
  - Provided all required steps clearly identified with different sections
  - Good cell separation (text, code, results, etc)
  - Tests on various examples easy to perform by the corrector
  - Report detailed enough for reproducibility
- Heading (-3 if missing)
  - Group number + student names + student numbers
  - Work split (provide information about who did what for the assignment, explain the work split between the 2 students as to know who was responsible for what)
- Dataset description (3 points)
  - Dataset name, author, purpose (what was it made for)
  - Shape: how many rows and columns
  - A list of the features + descriptions (what do they represent and are they categorical or numerical)
  - Explanation for choice of that dataset (if you choose the ones suggested or other ones)
- Data preparation (4 points)
  - Appropriate EDA has been included with a few tests described
- Study 1 – Similarity Measures (8 points)
  - Correctly uses the similarity measures
  - Program a text-based similarity measure
  - Shows the five requests + answers
- Study 2 – Clustering (8 points)
  - Correctly implements the clustering algorithms (KMeans, DBScan)
  - Tested the required number of variations (different parameters)
  - Showed and analysed resulting graphs (visual inspection)
- Study 3 – Content-Based Filtering (8 points)
  - Describe their similarity heuristics (domain justified)
  - Correctly programmed and explained the heuristics
  - Correctly used the heuristics in content-based filtering
  - Showed results of 3 requests with 2 heuristics + discussion
- Study 4 – Collaborative Filtering (12 points)
  - Correctly built the R matrix
  - Correctly performed the Matrix Factorization
  - Tested with different latent dimensions
  - Showed results of 3 requests (simulated as if they came from 3 different users)
  - Correctly performs/shows quantitative evaluation with MSE
  - Correctly performs/shows quantitative evaluation on relevance (P@K, MRR)

- References (-3 if missing)
  - For any part of your code taken from a web site (even a tutorial site or stackoverflow), you must provide the reference to it.
  - If part of your code was generated using Generative AI, please list the tool as well as the queries performed for each code snippet used.
  - Any theory/algorithms found in books, slides, tutorials that you used should be referenced.

Presentation/Demo (12 points)

- Understanding of the work
  - Students show a good understanding of the work they have done.
- Clarity of explanations
  - Students clearly express themselves using appropriate terms to describe what they have done.

QUESTIONS

- You can ask your questions within the assignment topic of the discussion forum on Brightspace.
- To make sure you get answers on the forum on a regular basis, the 3 TAs will share the days to answer your question. Notice that they are NOT available on weekends.

	Forum Monitoring						
	Monday	**Tuesday**	**Wednesday**	**Thursday**	**Friday**	**Saturday**	**Sunday**
Morning (before noon)	Svetlana	Ángel		Ángel	Svetlana		
Evening (before 9pm)	Shavbo	Shavbo	Svetlana	Shavbo	Ángel		

If you need to reach the TAs by email, you should use the following emails:

- Shavbo: ssale038@uottawa.ca
- Svetlana: sesina@uottawa.ca
- Ángel : avale073@uottawa.ca